

Improving Foggy Scene Segmentation via Teacher–Student Self-Training

Nguyen Phi Anh¹, Nguyen Thac Cuong¹, Nguyen Van Dai¹,
Pham Khac Tiep¹, Hoang Quoc Viet¹

^{1*}University of Engineering and Technology, Vietnam National
University Hanoi.

Contributing authors: 23020009@vnu.edu.vn; 23020018@vnu.edu.vn;
23020036@vnu.edu.vn; 22022560@vnu.edu.vn; 22021129@vnu.edu.vn;

Abstract

The segmentation of semantic images in outdoor foggy environments is being severely affected by the sudden change in the weather domains from the source domain (clear weather, labeled) to the target domain (foggy weather, typically unlabeled). The paper investigates a simple Teacher Student self-training method inspired by FixMatch/UniMatch-lite [2, 3], whereby an EMA teacher generates pseudo-labels from weakly-augmented views, and the student is trained on strongly-augmented views while applying pixel-level confidence thresholding to discard unreliable pseudo-labels. Experiments are carried out on the Foggy Zurich and Foggy Driving datasets [5, 6], while an additional clear control set (Lindau) is also used for performance reporting in order to monitor the possible negative transfer effect. Source code: <https://github.com/Anhnguyen0812/CVDHD>

Keywords: Foggy Segmentation, EMA, Self-training

1 Introduction

Semantic segmentation typically performs well when the weather is clear and the scene looks familiar. The model can rely on patterns it has already seen many times. However, when fog appears, it causes some damage to model and image, such as visibility to drop, contrast to be lower, and objects to look strange.

Now, most models are trained on clean images with labels. Because of this, models trained on clear-weather images often perform poorly when applied to foggy conditions. The main problem is not only the fog itself, but the gap between the training domain and the real testing domain.

One possible way to reduce this gap is by using self-training, although this approach has some risks. In this report, a Teacher–Student framework is used, inspired by FixMatch and UniMatch-lite [2, 3]. The idea is relatively simple. A teacher model, updated by exponential moving average, generates pseudo-labels from weakly augmented images, which still look close to the original data. The student model then learns from strongly augmented images using these pseudo-labels. Pixel-level confidence thresholding is used so that uncertain regions can be ignored, instead of forcing the model to learn from noisy labels.

The method is evaluated on the Foggy Zurich and Foggy Driving datasets [5, 6] and the Lindau - a clear-weather dataset, are also reported. This work aims to check whether adapting to foggy data may reduce performance on clear scenes, which can happen in some cases.

In this report, we use FIFO [1] as a strong baseline for semantic segmentation in foggy scenes. Because it includes fog-related filtering and extra training objectives. However, in real experiments, the performance is not always stable. It can change a lot depending on how long the model is fine-tuned and which checkpoint is chosen. There is also another issue. If the pseudo-labels are of low quality, the model may learn wrong patterns and repeat its own mistakes. On the other hand, if the model adapts too much to foggy data, its performance on clear scenes may decrease slightly. In real-world driving scenarios, where weather changes gradually, this trade-off is difficult to avoid.

Contributions

This report has two contributions. The first is that it proposes a Teacher–Student training pipeline for foggy semantic segmentation based on FixMatch and UniMatch-lite [2, 3]. The pipeline uses an EMA teacher, weak-to-strong augmentation, and confidence-based pseudo-label selection during training. The second contribution is that pixel-level confidence masking with a label is used to filter out pseudo-labels that are not confident enough, so the model is less to reinforce its own wrong predictions during learning.

Structure.

Section 2 shows related work, Section 3 reviews the background, Section 4 details the proposed approach, Section 5 shows the setup and research questions, Section 6 discusses the results and threats to validity, and Section 7 concludes the report.

2 Related Work

There are three related works to this report, includes: domain adaptation for segmentation (feature/output alignment between the source and target domains); self-training and pseudo-labeling; and Teacher–student with EMA combined with weak-strong augmentation. Domain adaptation reduces the distribution gap between domains, but under heavy fog conditions, visual signals in images are severely distorted, making

alignment no longer effective. Self-training and pseudo-labeling enable the exploitation of unlabeled target-domain data through model predictions, which is not exploited in the FIFO paper [1]. However, the model may reinforce its own incorrect predictions. A common solution to this issue is confidence-based pseudo-label filtering and the use of an EMA teacher. In Teacher–Student models with EMA and weak/strong data augmentation, the teacher is updated using the Exponential Moving Average, and the student learns from weakly and strongly augmented data. The method proposed in this report plays a role as an end-to-end pipeline, including: pixel-wise self-training with confidence masking, EMA teacher, supervised source anchor to reduce negative transfer, and additional Teacher–Student consistency loss to improve robustness.

3 Background

3.1 Semantic segmentation and mIoU

Semantic segmentation has the task of assigning a class label to each pixel in an image. The evaluation metric used for comparison in both the referenced paper and this report is the mean Intersection-over-Union mIoU. Example: IoU of the *car* class is computed as the area of overlap between the region predicted as car by the model and the corresponding ground-truth region, divided by the area of their union. In multi-class segmentation, mIoU is obtained by averaging the IoU values over all classes. Explaine in the formula below, TP_k denote the number of correctly predicted pixels of class k , FP_k denotesthe number of false positive pixels of class k , and FN_k denote the number of miss pixels of class k :

$$IoU_k = \frac{TP_k}{TP_k + FP_k + FN_k}, \quad mIoU = \frac{1}{K} \sum_{k=1}^K IoU_k. \quad (1)$$

3.2 Cross-entropy (CE)

With logits z , and a probability distribution resulting from the softmax function $p = \text{softmax}(z)$, together with the true label y , the cross-entropy loss is defined as:

$$CE(p, y) = -\log(p_y). \quad (2)$$

For semantic segmentation, the cross-entropy loss is taken over all valid pixels. And during loss computation, an *ignore label* is used to omit certain pixels.

3.3 Teacher–Student Self-Training, EMA, Confidence Thresholding

Teacher-student self-training uses a teacher model to create pseudolabels for the unlabeled data and trains the student to conform to these pseudolabels. Teacher of the exponential moving average EMA updates from the student parameters:

$$\phi \leftarrow \alpha\phi + (1 - \alpha)\theta, \quad (3)$$

$\alpha \in [0, 1)$ as the EMA decay rate.

4 Proposed Method

4.1 Notations and Inputs/Outputs

There are some notations, including: student model f_θ , teacher model f_ϕ ; source domain data with labels (x_s, y_s) , target domain data without labels x_t , weak augmentation a_w ; strong augmentation a_s , pixel-level pseudo-label $\hat{y}$ and confidence mask m ; threshold τ . Inputs: source domain data with labels (Cityscapes CW/SF [4]), target domain data without labels (Foggy Zurich / Foggy Driving [5, 6]), segmentation backbone (RefineNet-lw101 [7], as in FIFO [1]), hyperparameters. Outputs: The best student checkpoint.

4.2 Pipeline

A supervised source stream - for preserving core knowledge and a target self-training stream - for adapting to fog are used in our method. There are many iterations, and each iteration:

1. Sample labeled source minibatch (x_s, y_s) and unlabeled target minibatch x_t .
2. Two versions of the target data are generated, including: a weakly augmented view $x_{t,w} = a_w(x_t)$ and a strongly augmented view $x_{t,s} = a_s(x_t)$.
3. EMA teacher to $x_{t,w}$, export $\hat{y}$ and confidence mask m with threshold τ .
4. The student model is updated (minimizing the combined losses from the source and target streams).
5. Then, the teacher is updated using EMA.

4.3 Source stream: supervised loss

The student takes part in this stream, using fully labeled data from the source domain to predict the probability distribution $p_s(x_s)$, and minimize the cross-entropy loss. The goal is to keep the fine-tuning process stable and avoid catastrophic forgetting.

$$L_{src} = \text{CE}(p_s(x_s), y_s). \quad (4)$$

4.4 Target stream: FixMatch/UniMatch-lite with pixel-wise pseudo-labels

The target stream aims to use unlabeled foggy domain to help the model adapt to foggy conditions, while controlling noise and avoiding wrong self-learning. The EMA teacher takes weakly augmented images as input and produces a probability distribution for each pixel $p_t(x_{t,w})$ (softmax over classes). For each pixel, $\hat{y}$ is the class that the teacher believes is most likely, and c^* is the confidence of that prediction, which is used as the pseudo-label. The confidence mask m , with threshold τ , is used to filter out unreliable pixels.

$$\hat{y} = \arg \max_c p_t^c, \quad c^* = \max_c p_t^c, \quad (5)$$

$$m = \mathbf{1}[c^* \geq \tau]. \quad (6)$$

Student receives the strongly augmented image $x_{t,s}$, makes predictions, and computes the loss using only pixels where $m = 1$. Pixels with $m = 0$ are given an `ignore_label`, so they are ignored in the segmentation loss and not included in the CE average.

$$L_{tgt} = \frac{1}{\sum m} \sum m \cdot \text{CE}(p_s(x_{t,s}), \hat{y}). \quad (7)$$

Consistency loss (optional).

This loss encourages the student and teacher to give similar predictions on confident pixels.

$$L_{consistency} = \frac{T^2}{\sum m} \sum m \cdot \text{KL}(p_t(x_{t,w}) \parallel p_s(x_{t,s})). \quad (8)$$

4.5 Overall objective

$$L = L_{src} + \lambda_{pl} L_{tgt} + \lambda_{con} L_{con}. \quad (9)$$

4.6 Tracking negative transfer (only use in report)

This part is added to track whether the model suffers performance degradation under clear-weather conditions.

$$\Delta \text{Lindau} = \text{mIoU}_{\text{Lindau}}^{(our)} - \text{mIoU}_{\text{Lindau}}^{(FIFO)}. \quad (10)$$

5 Experimental Results

5.1 Datasets, baseline, and metric

This report is compared with FIFO [1] (the baseline) on three foggy datasets: Foggy Zurich (FZ) [5], FoggyDriving(FD) [6], FoggyDrivingDense (FDD). They are fully annotated. We also use an unseen clear weather dataset - Lindau. The main metric is mIoU.

5.2 Training configuration

In our experiments, we employed the Teacher-Student setup as discussed in Section 4. Our student is RefineNet-LW101 [7], similar to the FIFO setting [1]. Initially, we attempted to replicate the FIFO baseline completely, but we were not able to achieve the performance reported due to some unknown issues. In order to be sure that we are comparing against a strong baseline, we fine-tuned starting directly from the FIFO final checkpoint. We also employ a restrained SAFE recipe to stabilise the model

Table 1 Comparison FIFO baseline and Teacher-Student Self training (mIoU)

Method	FZ	FDD	FD	Lindau
FIFO(Baseline, final)	48.41	48.93	50.71	64.75
Ours(final run)	49.70	49.75	50.99	64.99

since our batch size is small. In our experiments, we set the batch size to 1 and use gradient accumulation (iter-size 4), and we often reviewed intermediate checkpoints. Unless said otherwise, we use an EMA teacher (high α), a confidence threshold τ , and optional consistency loss.

5.3 Research questions

- **RQ1 (Performance):** We verify the improvement in foggy mIoU of the newly proposed method over FIFO and also check if the Lindau drop constraint (≤ 0.2) is satisfied.
- **RQ2:** What is the impact of setting and scheduling τ (and its scheduling) on the ratio of selected pixels and mIoU respectively?
- **RQ3 (Ablation):** We study the role of each part: EMA, threshold scheduling, the consistency loss, and also the source branch.
- **RQ4 (Resources):** Is there a drastic increase in VRAM and runtime overhead compared to the baseline?

5.4 Answer to RQ1: Overall performance

Overview.

We compare the FIFO baseline with the final Teacher-Student self-training model and check the Lindau results to make sure that clear-weather performance is not reduced.

Overall results.

From the final FIFO baseline, the Teacher-Student self-training improves on all foggy datasets and maintains clear weather dataset performance. On our final run, performance has improved on all datasets by +1.29 mIoU on Foggy Zurich; +0.82 on Foggy Driving Dense; +0.28 on Foggy Driving; and Lindau increases by +0.24 (Table 1). The negative transfer constraint is ensured ($\Delta\text{Lindau} \geq -0.2$).

Finding summary.

[Finding-RQ1] Teacher-Student self-training significantly improved foggy mIoU over the FIFO baseline and at the same time satisfied the negative-transfer condition ($\Delta\text{Lindau} \geq -0.2$).

5.5 Answer to RQ2: Effect of τ and scheduling

The report presently applies a constant confidence threshold τ for pseudo-label masking. We have postponed the exploration of threshold scheduling (beginning from a

Table 2 Ablation study on Teacher-Student self-training (best SAFE checkpoint at 500 steps, mIoU).

Setting	FZ	FDD	FD	Lindau
FIFO (Baseline, final)	48.41	48.93	50.71	64.75
FixMatch	48.75	49.08	50.81	64.90
FixMatch without EMA	48.75	49.08	50.81	64.91
FixMatch without Threshold mask	48.75	49.11	50.82	64.90
FixMatch + consistency	49.01	49.49	50.97	64.83

high one and gradually lowering) to future work. Such an experiment should track the ratio of selected pixels over time to assess the stability.

5.6 Answer to RQ3: Ablation study

We have performed an ablation study on Teacher-Student self-training - gradually removed the key components to investigate the contribution of each component to the mIoU enhancement.

In all settings, we use the same SAFE fine-tuning strategy and present the best SAFE checkpoint after 500 training steps.

The main takeaway from our ablation study is that the consistency loss is the component that impacts the performance the most. When it is removed, the performance drops by -0.56 mIoU on FDD and -0.60 mIoU on FZ. On the other hand, dropping the confidence thresholding mechanism or turning off the EMA teacher hardly affects the performance under the SAFE setting.

5.7 Answer to RQ4: Resource usage

Considering the computational resources, the proposed method is more demanding than the original FIFO one. The additional forward pass of the EMA teacher on the weakly augmented images accounts for most of this overhead.

The pseudo-label generation and the consistency loss also bring some extra computation load. The EMA parameter update is low in terms of computation. GPU memory is also slightly increased due to the storing of teacher outputs, the application of strong data augmentations, and the rehearsal of the teacher forward pass.

6 Discussion and Threats to Validity

6.1 Discussion

We propose to improve the model performance: (i) setting a confidence threshold to select only high-confidence pixels for the next training cycle; (ii) using strong data augmentation to improve data generalization, which helps the model adapt better to real situations and testing; and (iii) applying a teacher-student design (with an EMA teacher) to prevent the model from forgetting what it learned from the previous data domain, thus avoiding unwanted performance drops.

In addition, during fine-tuning, special care is needed, especially for short training runs. A learning rate that is too high can make the model unstable, cause a fast drop in mIoU, and lead to model drift. With SAFE fine-tuning, this problem can be reduced, but it usually requires more training steps.

6.2 Threats to Validity

Internal validity.

The results can be affected by the choice of hyperparameters (e.g., τ , α , λ), the frequency of model snapshots, and random initialization. Therefore, fair comparisons require that both baseline and proposed approaches are evaluated under the same experimental settings.

External validity.

The conclusions drawn from experiments under foggy conditions may not directly apply to other scenarios such as rain or nighttime. In addition, the results may change when using different network backbones or foggy datasets that contain different biases.

Construct validity.

Using mIoU alone may hide performance differences on rare or difficult classes. When possible, it is recommended to additionally report per-class IoU or results on a selected set of challenging categories.

7 Conclusion

This report proposes an improved method based on FIFO [1], that is integrating FixMatch/UniMatch-lite self-training [2, 3] with an exponential moving average (EMA) teacher, confidence-based pseudo-label filtering, and (an optional) teacher-student consistency loss. Experimental results show that the proposed approach improves mIoU on the foggy domain, while still effectively controlling negative transfer. Several future directions are suggested, including applying data augmentations (such as flipping and cropping images), more tightly integrating FIFO’s fog-pass filtering module, and incorporating additional auxiliary modules.

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